Jacques Attinger

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Education

University of Chicago — Chicago, IL
B.S. in Mathematics and Computer Science, GPA: 3.82/4.0 — *Expected May 2028*

- **Relevant Coursework**: Data Structures and Algorithms, Data Analysis With Python, Real Analysis, Linear Algebra, Probability and Statistics

Work Experience

Hemut (YC X25) — Fresno, CA
Machine Learning Engineer — *Jun 2026 – Present*

- Built a **RAG** pipeline (Qdrant, OpenAI, Celery) that auto-bids **1000+ lane** RFP spreadsheets in parallel against historical shipment data, cutting processing time by **over 90%**
- Automated accounts receivable for a trucking TMS (Python, FastAPI, Node.js, PostgreSQL/SQL, React/TypeScript) via LLM-vision OCR document autofill and factoring exports to Excel/PDF, cutting workflow time by **over 80%**
- Deployed multi-tenant services to **AWS ECS** with Docker, automated testing, and GitHub Actions CI/CD as a forward-deployed engineer, turning weekly customer feedback into shipped features for fleets of **100+ trucks**

Argonne National Laboratory — Chicago, IL
Machine Learning Research Intern — *Jun 2025 – Aug 2025*

- Integrated **SLADS-Net** (deep-learning dynamic sampling) from research literature with SEM control software via the PyPhenom API, increasing image reconstruction accuracy by **21%** over random sampling methods
- Built an autonomous SEM image analysis pipeline (**OpenCV**, NumPy) that acquires X-ray spectra for **500+ particles** per run without human input; adopted by multiple Argonne research groups
- Deployed a **Convolutional Neural Network (CNN)** (PyTorch) to classify energy-dispersive X-ray spectra from a Scanning Electron Microscope (SEM) into elemental classes, reducing data acquisition times by **up to 90%**
- **Publication**: Co-author, Towards autonomous imaging workflows using scanning electron microscopy for materials research, *MRS Communications* (accepted, 2026)

University of Chicago Department of Physics — Chicago, IL
Undergraduate Researcher — *Feb 2025 – May 2026*

- Designed a safety interlock system linking lab sensors with control software to automatically shut down quantum manufacturing systems on overheating or underpressurization, saving thousands of dollars in equipment
- Developed a Tkinter **GUI** app to store and visualize real-time vacuum chamber pressure/temperature data
- Engineered an interactive PyQt GUI and communication layer to control a machine-vision camera and pyrometer, enabling synchronized thermal and visual data acquisition for Reinforcement Learning model training

Princeton University Department of Mechanical & Aerospace Engineering — Princeton, NJ
Computational Research Intern — *Jun 2023 – Aug 2023*

- Ran Cantera e-fuel oxidation simulations to find experiment–model gaps, improving Dimethoxymethane reaction pathways
- **Publication**: Co-author, Dimethoxymethane low- and intermediate-temperature oxidation up to 100 atm, *Proc. Combust. Inst.* (2024)

Iowa State University Poly Cy Lab — Ames, IA
Researcher — *Jan 2023 – May 2023*

- Operated extrusion and injection-molding manufacturing technologies to aid in biobased polymer production
- **Publication**: Co-author, Fully Atom-Efficient Solvent-Mediated Biopolymer Manufacturing: A Base Case Illustrated with Macromolecular Surfactants Tailored to Stable Polymer–Water Interfaces, *ACS Applied Materials & Interfaces* (2024)

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS

Tools: PyTorch, LangChain, LangGraph, FastAPI, React, Node.js, PostgreSQL, Qdrant, Docker, AWS (ECS, S3, SQS), Celery, GitHub Actions, CI/CD, Git, GitHub, Linux, NumPy, OpenCV, Claude Code, Codex, Vercel